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Special Section:
Disturbance Hydrology: Assessing the Immediate and Long-Term Impacts of Abrupt Landscape Changes on Hydrologic Processes and Function

Key Points:
- Landslides often remobilize or recur in the location of previous slope failures, which complicates hazard assessments
- Continuous monitoring of a control hillslope and recent landslide reveal event and seasonal differences in hillslope hydrology and stability
- Landslide disturbances reduce hydrologic connectivity and storage, which both enhances and prolongs susceptibility to further sliding

Supporting Information:
- Supporting Information S1

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Hydrologic Impacts of Landslide Disturbances: Implications for Remobilization and Hazard Persistence

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Abstract Landslides typically alter hillslope topography, but may also change the hydrologic connectivity and subsurface water-storage dynamics. In settings where mobile materials are not completely evacuated from steep slopes, influences of landslide disturbances on hillslope hydrology and susceptibility to subsequent failures remain poorly characterized. Since landslides often recur at the site of previous failures, we examine differences between a stable vegetated hillslope (VH) and a recent landslide (LS). These neighboring hillslopes exhibit similar topography and are situated on steep landslide-prone coastal bluffs of glacial deposits along the northeastern shore of Puget Sound, Washington. Our control hillslope, VH, is mantled by a heterogeneous colluvium, supporting a dense forest. In early 2013, our test hillslope, LS, also supported a forest before a landslide substantially altered the topography and disturbed the hillslope. In 2015, we observed a clay-rich landslide deposit at LS with sparse vegetation and limited root reinforcement, soil structures, and macropores. Our characterization of the sites also found matrix porosity and hydraulic conductivity are both lower at LS. Continuous monitoring during 2015–2016 revealed reduced effective precipitation at VH (due to canopy interception), an earlier seasonal transition to near-saturated conditions at LS, and longer persistence of positive pore pressures and slower drainage at LS (both seasonally and between major storm events). These differences, along with episodic, complex slope failures at LS support the hypothesis that, despite a reduced average slope, other disturbances introduced by landsliding may promote the hydrologic conditions leading to slope instability, thus contributing to the persistence of landslide hazards.

Plain Language Summary The most deadly landslides in recent U.S. history resulted from reactivation of existing landslides. Motivated by these tragedies and other landslide-related losses, we investigated why recent landslides may be less stable than similar hillslopes nearby. We collected observations and monitoring data from two neighboring hillslopes sharing similar geology and steep topography, near Seattle, Washington; one hillslope where a landslide occurred recently and a second that was stable and forested. We found that this second hillslope remained stable in part because roots reinforce the soils, but also because the vegetated soils drain efficiently following major rainstorms. In contrast, the existing landslide on the other hillslope disrupted vegetation and altered the soil properties, reducing drainage efficiency. As a result, the landslide materials became wetter earlier in the winter rainy season than the soils of the neighboring vegetated hillslope and also stayed wetter longer after the rains ended. The combination of reduced root reinforcement and the persistence of wetter soil promoted repeated slope failures at the landslide hillslope. These new insights into the "self-driving" factors contributing to landslide recurrence can be used to inform residents in landslide-prone areas and improve landslide hazard assessments to help reduce landslide-related losses.

1. Introduction and Background

Landslides are more likely in regions where they have occurred in the past. This generality has facilitated the development of susceptibility maps using the presence of previous landslides along with terrain attributes such as geology, vegetation soils, topography, tectonic, and climatic factors (e.g., Burns et al., 2016; Kirschbaum et al., 2016; Radbruch-Hall et al., 1982; Shahabi & Hashim, 2015). Anthropogenic disturbances such as timber harvesting and forest roads alter root reinforcement, topography, and subsurface properties, which often increases landslide susceptibility for a given location or area (Mirus et al., 2007; Montgomery et al., 2000; Swanson & Dyrness, 1975; Wemple et al., 2001). However, landslides themselves may also introduce disturbances, which are rarely considered in typical hazard assessments, yet may influence the potential for landslide reactivation or recurrence in the same location. In situations where the failed material is completely evacuated, by a debris flow for example, landslides generally recur in that same location, but only after weathering and deposition of material from upslope once again provide a sufficient supply of mobile sediment (Imaizumi et al., 2015; Jakob et al., 2005; Reneau & Dietrich, 1991). However, following lower mobility landslides, failed materials are not completely evacuated from the steeper slopes, yet the average slope from scarp to toe is generally reduced. Therefore, based on topography alone, one might expect that the initial occurrence of such landslides would translate hillslope materials from a previously unstable state into one of balanced equilibrium. This is not always the case, and some of the most deadly landslides in the U.S. highlight the tragic reality of catastrophic reactivation of recent or historical landslide deposits.

On 10 January 2005, in La Conchita, California, material from a prior landslide in 1995 remobilized into a highly fluid, rapidly moving debris flow, resulting in 10 fatalities and severe damage to 36 homes (Jibson, 2005). On 22 March 2014, in Oso, Washington, the 2006 Hazel landslide reactivated and failed; material traveled rapidly as a debris-avalanche flow, resulting in 43 fatalities and extensive and costly damages to private property and state highways (Iverson et al., 2015). On 25 May 2014, at West Salt Creek near Collbran, Colorado, a massive rock avalanche initiated from a preexisting rockslide, resulting in three fatalities and a runout deposit that encroached to within several meters of an active oil and gas production well (Coe et al., 2016).

A more widespread example of landslide recurrence is associated with heavy rainfall during the winters of 1996 and 1997, which impacted landslide-prone regions of Puget Sound, Washington, just 40 km southwest of Oso in similar glacial deposits. Along the 32-mile railway corridor between the cities of Seattle and Everett, at least 132 landslides resulted from those winter storms (Baum et al., 2000). Baum et al. (2000) compared their map of those recent landslides with prior geologic mapping in the area (Smith, 1975, 1976) and found that failures occurred in 9 out of 10 late Pleistocene landslide deposits and in 16 out of 21 landslides that were mapped as active in the 1970s. Thus, in addition to many new landslides a significant fraction of the prior landslides reactivated in 1996–1997.

There are many possible explanations for landslide reactivation or recurrence in the same location. Steep convergent topography is known to contribute to the recurrence of landslides due to pore-pressure buildup, which evacuates colluvium-filled bedrock hollows (Ebel et al., 2008; Montgomery & Dietrich, 1994; Wu & Sidle, 1995). However, in settings without steep, strongly convergent bedrock topography, such as the relatively planar coastal bluffs along Puget Sound, the influence of lateral subsurface flow convergence is limited. Undermining the toe of previous landslides by wave action or bank erosion is another possible explanation for landslide reactivation (Collins & Sitar, 2008; Minard, 1983; Pettit et al., 2014), but along the Seattle-Everett corridor engineering structures for the railway have mitigated the influence of coastal processes on sediment removal. Therefore, further explanations for landslide reactivation and recurrence in a variety of different physiographic settings should be explored.

Advanced physically based modeling of debris flow runout demonstrates that very slight changes in subsurface properties such as initial porosity can drastically influence runout mobility, which explains why Oso and similar landslides may exhibit limited movement during one failure episode and then fail catastrophically at a later date (Iverson & George, 2016). Similarly, the role of porosity and saturated hydraulic conductivity in controlling the hillslope hydrologic processes driving precipitation-induced landslides are conceptually well understood (e.g., Bogaard & Greco, 2016; Collins & Znidarcic, 2004; Lu & Godt, 2013; Sidle & Ochiai, 2006). In seismically active zones, ground motion from earthquakes can not only trigger landslides (e.g., Harp & Jibson, 1996), but can also lead to consolidation of porous media (Rutter et al., 2016), which may explain why those slopes that did not fail during an earthquake are more susceptible to subsequent precipitation-induced landslides (Keefer, 1994). However, little is known about the changes in subsurface properties resulting from prior landslide disturbances, and how those changes may subsequently affect hillslope hydrology, slope stability, and remobilization. Abrupt disturbances such as wildfire, deforestation, earthquakes or volcanic eruptions can alter porous media properties, resulting in critical changes to hydrologic storage capacity and flow path connectivity (Ebel & Mirus, 2014), but this phenomenon has yet to be fully explored in the context of landslide hazards. Here we investigate one possible explanation for the persistence of landslides in the Puget Sound area within the context of disturbance hydrology. We present empirical evidence that disturbances to vegetation and subsurface properties associated with shallow landslides may result in hydrologic conditions that promote further slope failures.

2. Field Site and Methods

We used a pair of hillslopes with similar characteristics to examine the surface and near-surface impacts of disturbances associated with shallow landsliding and their combined influence on hydrology and slope stability. Our neighboring hillslopes, VH and LS, are located on steep, landslide-prone coastal bluffs along the northeastern shores of Puget Sound, Washington, within the city of Mukilteo (Figure 1). Although several previous landslides along this Seattle-Everett railway corridor have been mitigated to prevent reactivation, and massive boulders have been placed below the railway bed to limit undermining from wave erosion, these two hillslopes above the railway have not been reinforced to prevent landsliding. The VH hillslope is our control for pre-landslide conditions, and is located only 0.5 km south of LS, which represents post-landslide conditions. Both hillslopes were instrumented with an array of soil moisture and pore pressure

Figure 1. Map of Puget Sound Washington, showing location of field site in Mukilteo. Hillshade inset shows 0.3 m DEM with location of hillslopes monitoring sites, LS and VH.

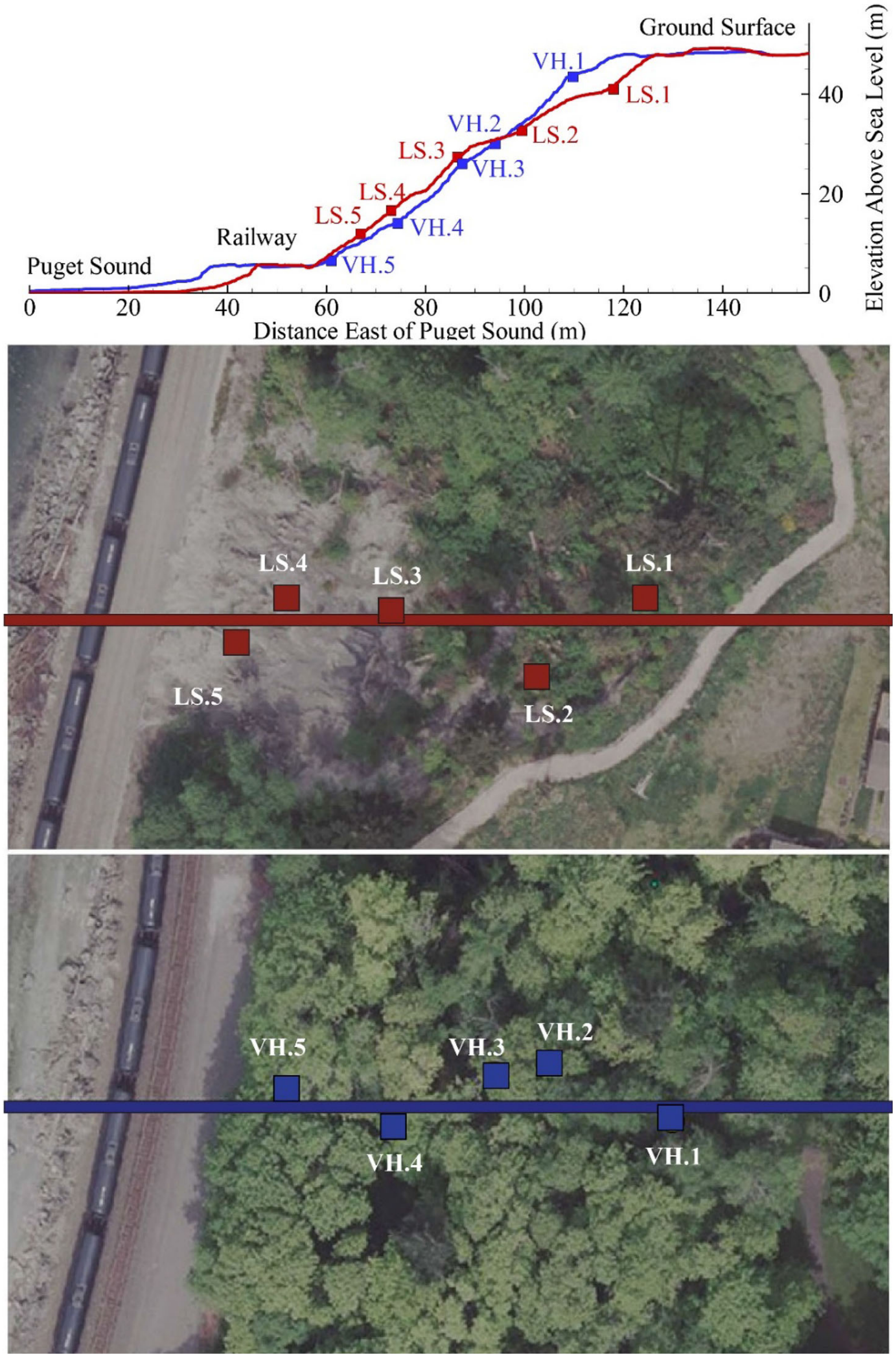

Figure 2. Topography and aerial imagery of LS and VH with locations of soil pits and monitoring instrumentation along transects.

sensors at five locations along longitudinal transects (Figure 2). Both hillslopes are situated below suburban developments, but neither displays any evidence of direct overland flow contributions from storm-water runoff, which is a recognized contributor to landslide initiation in this region.

While VH has remained stable during this study with no evidence of slope movement (through 21 May 2017) and supports a dense mixed forest canopy and understory vegetation, LS is on a landslide that failed in 2013, which visibly disturbed the vegetation (Figures 2 and 3). The headscarp at LS exposes a cross

Figure 3. Photographs illustrating differences in vegetation cover between VH and LS. LEFT: view facing downslope from near location VH.2 (Figure 2), with second author for scale and railway lines visible through the trees (photographed on 15 July 2016). RIGHT: view facing downslope from location LS.3 (Figure 2) with right lateral margin of landslide marked by the dense forest canopy; railway lines clearly visible at the base of the hillslope (photographed on 9 February 2016).

section of the undisturbed (pre-landslide) stratigraphy, which is capped by a 10 cm thick organic-rich soil horizon with abundant roots (see Mirus et al., 2016a). These observations suggest that prior to this initial failure in 2013, LS was vegetated much like VH with a similar near-surface soil profile. The two hillslopes exhibit the same total relief from the railway to the bluff crest (Figure 2) and share the same underlying geology of heterogeneous glacial till and outwash deposits (Booth et al., 2003; Minard, 1983). Despite this considerable local heterogeneity of the underlying glacial deposits that may influence the deeper groundwater flow system, we assume that—given the close proximity and general similarities between the two hillslopes—the VH hillslope is representative of the conditions prior to landsliding and that the primary differences in hillslope hydrology and stability observed at LS are directly or indirectly related to the landslide disturbance. This assumption is supported by the sequential response with depth observed at both seasonal and event time scales in our array of volumetric water content and pore-pressure sensors at both hillslopes (supporting information Figure S1) (Smith et al., 2017a), which suggests that largely vertical infiltration processes dominate any lateral groundwater inputs from upslope.

In this paper, we first examine the direct impacts of landslide disturbance through qualitative comparisons of the topographic cross sections, as well as soil structures and root distributions observed in soil pits excavated at each of the sites. We are also able to quantitatively compare the laboratory measured hydraulic properties of specimens collected from approximately the same depths at both sites. Second, we examine the indirect influence of the disturbance on hillslope hydrology by comparing time-series of sub-canopy rainfall measurements, as well as pore-pressure, and volumetric water contents measured at similar depths for each of the sites. We consider differences between effective rainfall (accounting for canopy interception) and subsurface hydrologic response at the two sites during both seasonal and event time scales. For a more direct comparison of hillslope conditions, we converted the volumetric water contents into soil saturations using the maximum observed water content of each probe during saturated conditions, which we inferred from periods of measured positive pore pressures. We initially installed twice as many probes at LS to monitor hydrologic conditions at multiple depths and in exposed scarps of the underlying glacial deposits, but for the analyses presented here we only compare analogous probes placed at equivalent depths in the potentially mobile regolith at both LS and VH. Finally, we evaluate the physical and hydrologic differences between the two sites in the context of the further episodic slope failures during the 2015–2016 landslide season, which we documented using extensometers and time-lapse cameras. Comprehensive details about the methods for our laboratory testing and instrumentation for our continuous monitoring and landslide detection are reported elsewhere (Mirus et al., 2016a, 2016b; Smith et al., 2017a, 2017b), but a brief summary is also provided in the supporting information Table S1.

3. Results of Landslide Disturbances

3.1. Direct Impacts

3.1.1. Field Characterization

The primary qualitative differences we observe between the two hillslopes are related to the topography, vegetation patterns, and regolith. The differences in topography are relatively subtle; the LS exhibits a modest decrease in average slope between the bluff crest to the toe (from 35° to 32°), though the toe remains at largely the same steep slope while the upper slump blocks of the slide have resulted in two moderately sloped steps (Figure 2). In contrast, the differences in vegetative canopy are stark. The VH supports a dense, mixed forest with understory vegetation of ferns and shrubs, while the LS has a very sparse vegetation concentrated as some tilted bushes and small trees on the slump blocks, and a few ferns and patches of puzzle grass (*Equisetum*) near the toe of the landslide (Figures 2 and 3). The VH is mantled by approximately one meter of colluvium with an upper layer of sandy-silty duff underlain by slightly developed horizons of reworked clay-rich glacial deposits, with ample evidence of bioturbation, as well as considerable macropores and root density (Figures 4a and 4b). The near surface of the LS hillslope is composed of a landslide deposit with similar geologic materials, but no duff layer or no soil development, particularly at depths below 10 cm (Figures 4c and 4d). These differences confirm that the landslide has largely removed the

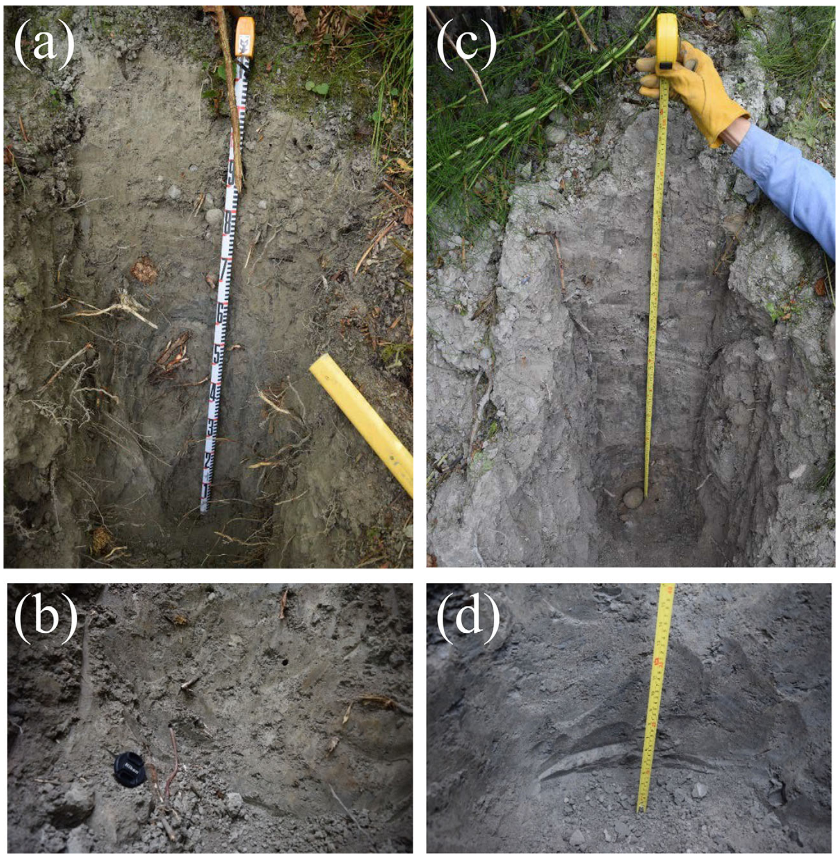

Figure 4. Photographs illustrating differences in soil structure, macropores, and root distribution. (a) soil pit at VH.4 with measuring tape for scale; (b) detail of soil pit VH.5 for 70–100 cm below land surface shown with 50 mm lens cap for scale (note large worm exiting from macropore); (c) soil pit LS.3 with measuring tape for scale; (d) detail of soil pit LS.5 for 80–120 cm below land surface shown with measuring tape for scale.

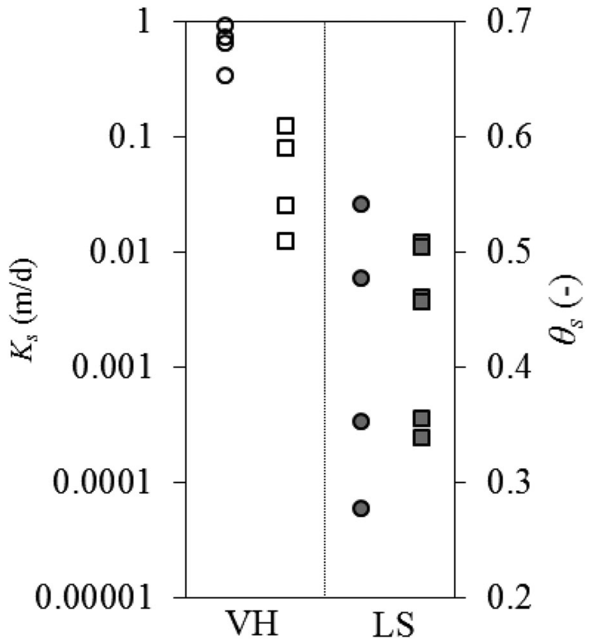

Figure 5. Lab measured saturated hydraulic conductivity, K_s (circles) and porosity, θ_s (squares), for VH (open symbols) and LS (filled symbols). Four specimens collected from VH were analyzed for K_s and θ_s; six specimens collected from LS were analyzed for θ_s and four for K_s (see Mirus et al., 2016a).

influence of vegetation at both the surface as well as within the subsurface, and also eliminated the obvious soil structures and preferential flow pathways that could promote lateral drainage of the hillslope.

3.1.2. Laboratory Measurements

Although both hillslopes are underlain by similar heterogeneous glacial deposits of clays, silts, sands, gravels, and boulders at depth (see Mirus et al., 2016a), the saturated hydraulic conductivity and porosity values of the core specimens collected from the base of the shallow soil pits at LS are clearly lower than those collected at similar depths at VH (Figure 5). Specifically, the geometric mean value of the saturated hydraulic conductivities is 0.622 m per d for VH and 0.001 m per d for LS, over two orders of magnitude lower. Similarly, the geometric mean value of the porosities is 0.56 for VH and 0.43 for LS, which is over 20% lower. These considerably lower values suggest that while the geologic materials are unchanged by the landslide disturbance, the hydraulic properties of the matrix are altered to reduce hillslope drainage and unsaturated storage capacity alike.

3.2. Indirect Impacts

3.2.1. Rainfall Monitoring

The continuous rainfall records for the 2015–2016 monitoring period (Figure 6) show that the timing and magnitude of storms are similar at both sites, but there are also notable differences due to the canopy interception effects at VH. The LS exhibits higher peak hourly rainfall intensities for the major storm events, but the timing of individual storms recorded by the tipping-bucket rain gages is slightly shifted between the two sites resulting in a seemingly random pattern of differences between hourly intensities recorded at the two sites (not shown). Plotting the daily differences between the two sites (Figure 6) reveals that for individual storms more rainfall is recorded at LS than VH. This trend in observed storm size also explains the 228 mm difference in cumulative rainfall between the two sites recorded during the entire 2015–2016 monitoring season (LS = 1160 mm, and VH = 932 mm). The shifts in timing recorded by the two tipping-bucket rain gages can also explain the episodic fluctuation between positive and negative daily differences recorded during the leaf-off period in January and February, when the cumulative difference between the sites remains fairly steady. This indicates that while canopy interception primarily dampens storm intensities in the early winter and early spring (i.e., December and March), the effect is to limit cumulative rainfall reaching the ground surface over the course of the year by about 20%, thereby strongly influencing the hillslope water balance.

3.2.2. Subsurface Monitoring

The continuous hydrologic response recorded during the 2015–2016 monitoring period (Figure 7) shows a distinct seasonal difference between the two hillslopes. The onset of positive pore-water pressures and near-surface soil saturations occurs in mid-November at LS and then in mid-December at VH. Similarly, the near surface begins drying out in the following April–May at the VH and continues steadily through the summer, whereas the decrease in pore pressures at the LS begins much later in May and both pore pressures and saturations at LS decrease relatively slowly. The shallow water-table develops in the upper slump block (LS.1b) at the beginning of November and is maintained at elevated levels through the end of March at which point it gradually recedes through the spring and early summer. In the piezometers installed in the headscarp composed of glacial deposits (LS.1a) and in the toe of the landslide deposit (LS.5a), measurements show that a water table first develops in early December and seems to respond only to particularly stormy periods rather than staying elevated for the entire season. This spatial variation in shallow water-table development and dynamics indicates that the upper portions of the landslide deposit saturate before the undisturbed glacial deposits above and the landslide runout deposits below, likely in response to the altered slump-block topography and hydraulic properties, rather than due to some unidentified deeper groundwater input from upslope. In contrast to the saturated zone response at LS, a shallow water table first develops at VH in late January, mostly at the base of the slope (VH.5), and then recedes in early April. Overall these seasonal differences in hillslope wetting and drying suggest that LS experiences a protracted "landslide season" since the hillslope remains closer to the elevated moisture conditions where landslides are possible for an extended period of time relative to VH.

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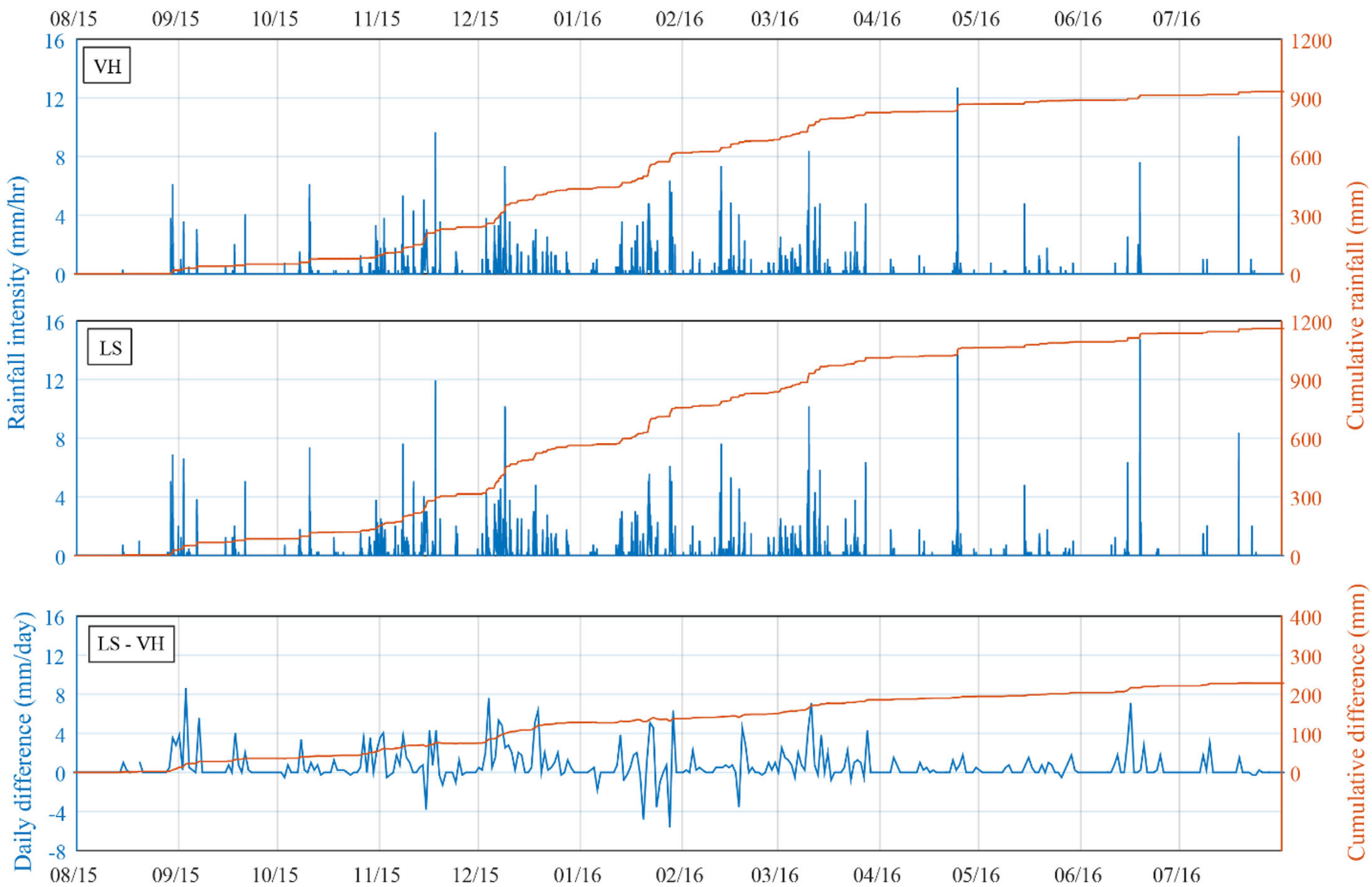

Figure 6. Hyetograph of measured rainfall intensity and cumulative depth, and differences for VH and LS during 2015–2016 monitoring.

In addition to the extended duration of the landslide season at LS, the two hillslopes also exhibit different hydrologic response to individual storm events. For example, during the major landslide-inducing storm sequence in January 2016, responses in piezometers screened over 2.7-3.0 m depths below land surface at both the toe and near the crest of each hillslope reveal some notable differences in the saturated zone drainage dynamics (Figure 8). In particular, the water levels in the slump block at the top of the landslide deposit (LS.1b) are the first to respond to the sequence of rainfall events, and water levels also remain elevated, approaching conditions indicative of the entire slump block becoming saturated. Nearer to the toe of the landslide deposit (LS.5a) water levels respond rapidly to the first major storm on January 13, but are slightly less responsive to subsequent storms, and although water levels remain elevated through the end of January, they do not approach completely saturated conditions in the toe of the landslide deposit. In contrast to these observations at LS, the water levels at VH show a more intuitive response to the same storm sequence (Figure 8). The lower piezometer (VH.5) only begins to respond after the peak response at LS occurs, and water levels then gradually increase until the heaviest series of storms starting on 21 January. Following this initially gradual increase in saturated thickness at the toe of the hillslope, the water table at VH rises more rapidly and with sufficient magnitude to also exhibit a small and brief response in the piezometer at the bluff crest (VH.1). Following the conclusion of the storm sequence at the end of January, the water table at the toe of the stable hillslope (VH.5) is considerably higher than at the toe of the landslide (LS.5a), but near the bluff crest the water table is lower (VH.1) than in the upper slump block of the landslide (LS.1b). These differences in saturated zone responses suggest that while at VH the water table follows the expected spatial pattern as a subdued replica of the topography (e.g., Alley et al., 1999), at LS the decreased hydraulic conductivity and porosity severely limits lateral downslope drainage from the upper slump block in particular.

The tensiometer responses during the same January storm sequence highlight further differences between the two hillslopes. Prior to the storm sequence, positive pore pressures had already developed at the mid-

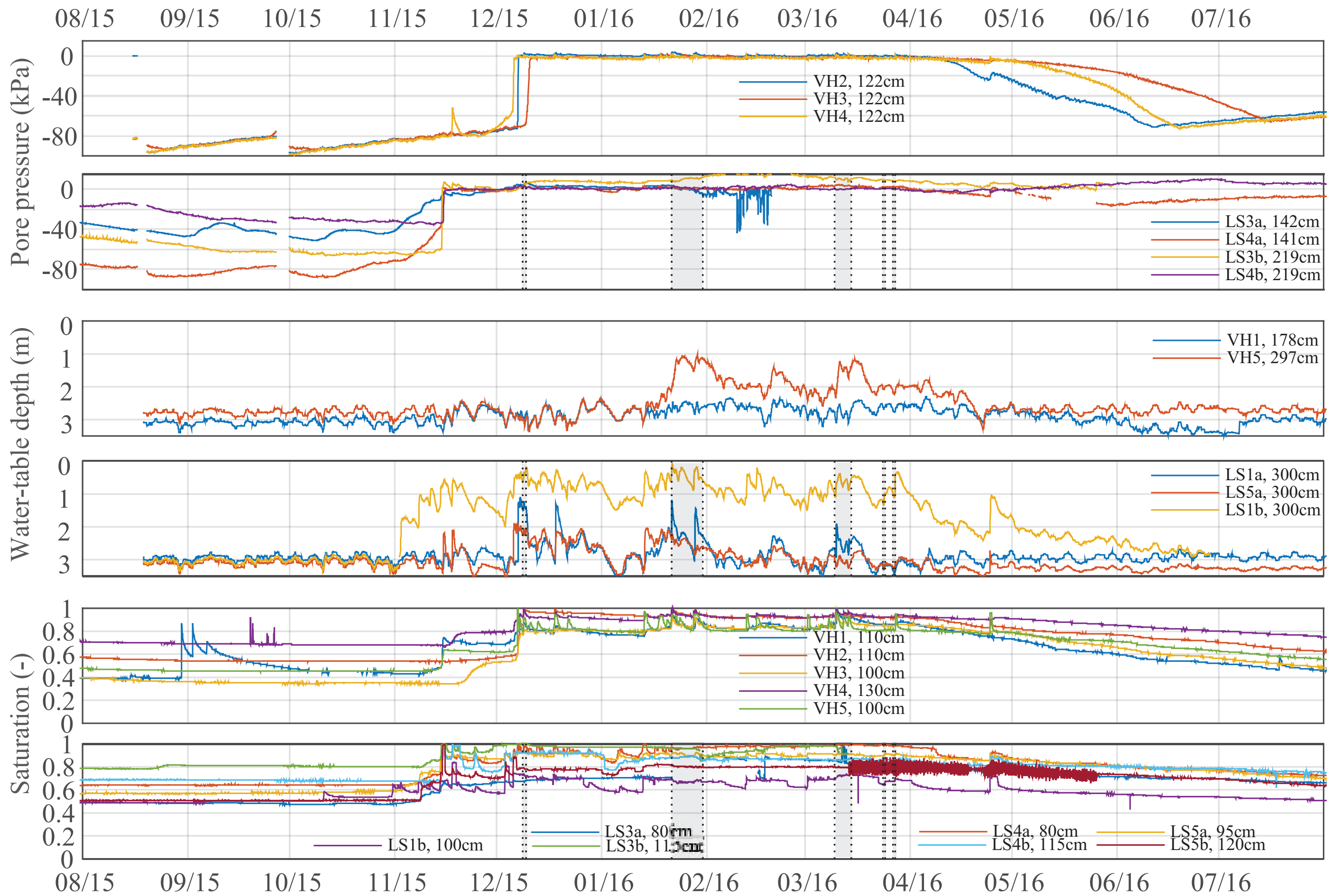

Figure 7. Measured pore-water pressures, shallow water-table fluctuations, and soil saturations for VH and LS during 2015–2016 monitoring. Shaded areas on the LS plots indicate timing of *known* landslide remobilization recorded by a time-lapse camera in the upper regions of the hillslope (LS.1 and LS.2, see Figure 2).

slope LS monitoring position (LS.3a), where pressures rapidly increased in response to the first rainfall event on 13 January and maintained elevated pore pressures even between storm events. This likely contributed to the landslide that occurred on 21 January at that location on the slope, which resulted in a combination of displacement and destruction of the probes (Figure 9 and supporting information Figure S1) (Smith et al., 2017a). The positive pore-pressure responses at LS.4a and VH.2 are quite similar, but the other two tensiometers at VH, located in lower and mid-slope positions (VH.3 and VH.4), recorded near-saturated conditions between rainfall events and only displayed positive pore-pressures during peak response to individual storms. Thus, while the two heterogeneous hillslopes exhibit a variety of responses, overall LS is wetter and drains more slowly, and the thicker saturated zone and slower drainage at LS continues for subsequent rainfall events (not shown). In general, the saturated zone responds more rapidly to storms and the near-surface maintains a higher degree of saturation and pore-pressures between storm events at LS as compared to VH, which contributes to the higher peak pore pressure that promotes the onset of landslide reactivation (see supporting information Figures S1–S3).

3.3. Landslide Recurrence

The episodic slope failures at LS during the monitoring period disrupted some electrical connections, and also damaged and displaced many of the shallower probes, thus jeopardizing the quality and consistency of our data in the latter part of the monitoring period (Figures 7 and 9). For example, the LS.3a pore-pressure sensor was dislodged and then broken during a series of slope failures, leading to erroneous measurements after 21 January (Figure 9); the electrical connection to the LS.3a water content sensor was subsequently disrupted, leading to noisy measurements from mid-March through the end of May when we

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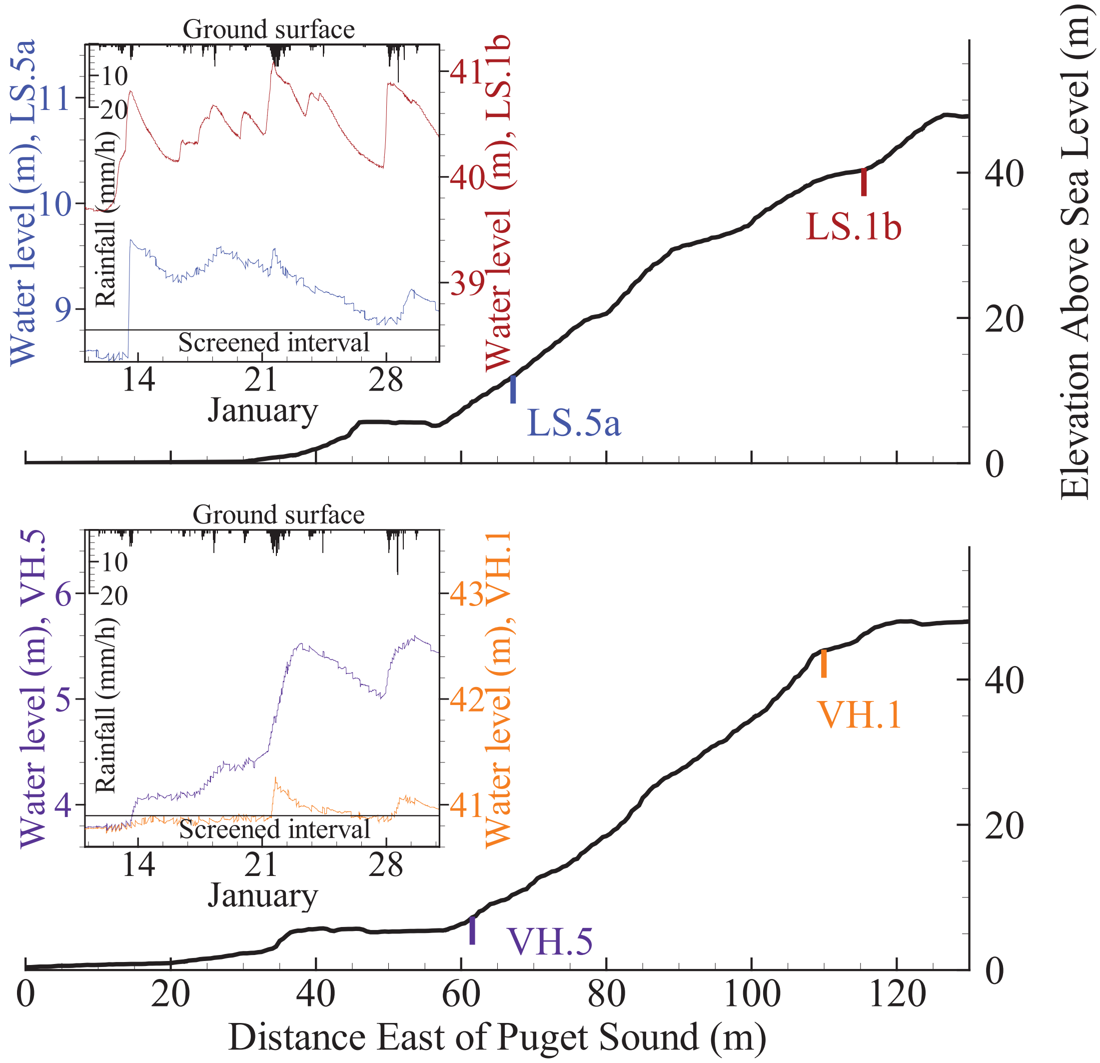

Figure 8. Transects showing locations and depths of comparable piezometers at LS and VH with insert graphs of water-level fluctuations in response to the landslide-triggering storm events in late January (see Figures 2 and 7).

repaired the connection (Figure 7). Furthermore, deposition of displaced regolith materials in the low slope position (LS.4) leaves some uncertainty of the effective probe depths in this vicinity following the slope failures in late January 2016. Regardless, despite these complications with data quality at LS following episodic slope failures, our comparison of the overall hydrologic response patterns at both seasonal and event scales reveals notable differences between LS and VH.

A combination of cable and laser extensometer monitoring and time-lapse camera images were used to document the timing of several of these episodic slope failures at LS, which are shown along with the hydrologic response data (Figure 7). In addition to the shallow translational movement documented by the camera in the upper portions of the slope near LS.1 and LS.2 (see Smith et al., 2017a), we also documented evidence of complex slope failures at various slope positions during site visits on 9 February and 26 May 2016, but with the exception of the failure at LS.3a on 21 January that can be identified by an abrupt change in pore pressures (Figure 9) we could not constrain the exact timing or even specific storms associated with most of these failures (see supporting information). Such complex failures included topples of intact glacial deposits from regressive failure along the headscarp (LS.1) and left-lateral scarp (LS.2), slides of smaller fragments of the same intact glacial deposits in the vicinity of the slump block benches (between LS.2 and LS.3), and translational slides and mudslides in lower portions of the toe slope (LS.3 and LS.4). Photographic evidence of these complex slope failures is provided in the supporting information Figures S4–S8.

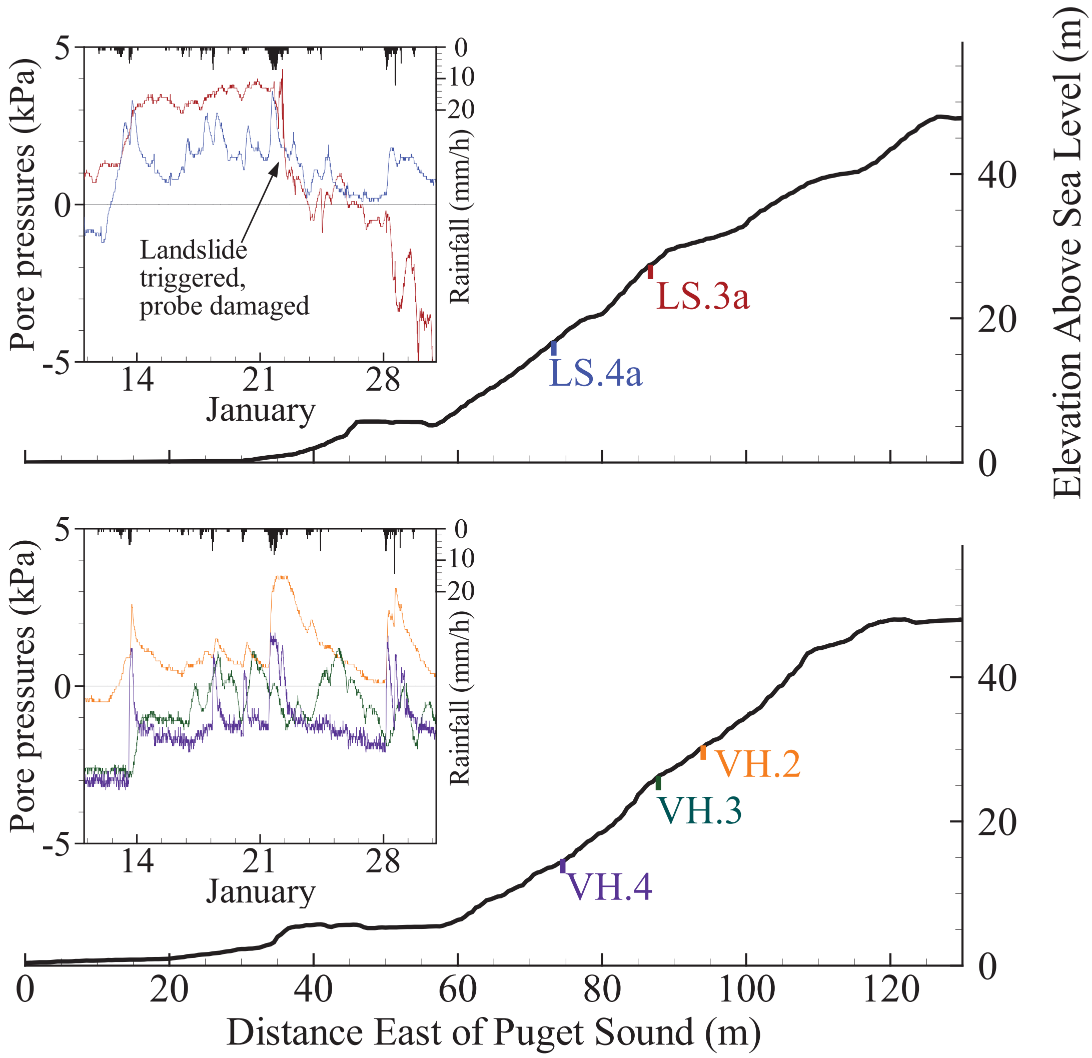

Figure 9. Transects showing locations and depths of equivalent tensiometers at LS and VH with insert graphs of pore-pressure dynamics in response to the landslide-triggering storm events in late January (see Figures 2 and 7).

4. Discussion

Abrupt landscape disturbances and the subsequent recovery can be examined in terms of changes in hydrologic connectivity and storage capacity, which can have important implications for water and nutrient fluxes, as well as landscape evolution (Ebel & Mirus, 2014). Here we further examine this concept of disturbance hydrology in the context of shallow landslide hazards in situations where the failed regolith is not completely evacuated by the slide. One of the primary challenges with investigating hydrologic disturbances is the lack of detailed pre- and post-disturbance monitoring data, which in the case of landslides is further complicated by the destructive nature of the disturbance. Even if it had been possible to identify the imminent failure of LS and instrument the hillslope prior to the initial landslide in 2013, our monitoring equipment sustained irreparable damage from the relatively minor landslide remobilizations observed in 2016. Thus instead of a truly paired approach with pre- and post-disturbance monitoring at one of the two hillslopes, our analysis hinges on the space-for-time substitution. We assume that the differences in topography, vegetation, and soil properties between our control hillslope (VH) and the landslide (LS) can be directly attributed to the disturbances associated with the slope failures, which initiated at the LS in 2013. Similarly, given the close proximity of these two otherwise similar hillslopes, the predominantly vertical infiltration processes, and the dearth of evidence for upslope water inputs from either storm runoff or deeper groundwater flow systems, we assume that any significant differences in rainfall and the corresponding near-surface hydrologic responses are the indirect result of these landslide-related disturbances. Over the course of an entire year, we monitored hydrologic state variables at multiple points along hillslope transects to determine representative measures of their dynamics and spatial distribution. Despite the necessary assumptions associated with a space-for-time substitution in heterogeneous hillslopes and the possible shortfalls of using a limited number of point measurements at only two hillslopes, our results suggest that landslide disturbances cause notable changes in vegetation patterns, subsurface properties, and topography, which in turn influence hillslope hydrology and stability.

Vegetation has an obvious and direct connection to slope stability because root strength contributes to an apparent cohesion that reinforces slopes (Hales et al. 2009; Roering et al., 2003; Schmidt et al., 2001; Sidle, 1992; Ziemer, 1981). Vegetation also exhibits indirect connections to slope stability via hillslope hydrology. For example, evapotranspiration controls antecedent wetness conditions that influence landslide susceptibility both during individual storm events and seasonally (Godt et al., 2006; Napolitano et al., 2015). Similarly, canopy interception is known to limit effective precipitation reaching the ground surface and reduce infiltration, which indirectly enhances slope stability (Keim & Skaugset, 2003; Reid & Lewis, 2009). Our tipping-bucket rain gages were installed at approximately 1 meter above the ground surface (see Smith et al., 2017a), so the differences between precipitation at LS and VH (Figure 6) reflect only the impacts of the forest canopy and not the understory vegetation. Thus, our measure of canopy interception effects is likely a somewhat conservative estimate. Furthermore, by comparing only two rain gages (one at each hillslope) we are unable to quantify any spatial variations in canopy interception, so considerable uncertainty remains in the effective interception across VH.

While we do not directly measure evapotranspiration (ET), it is reasonable to assume that due to the stark differences in vegetation average ET rates are lower for LS than for VH, which further contribute to wetter conditions observed year round at LS. Another vegetation-related hydrologic process that we were not able to consider is stemflow, which is very difficult to measure (Lewis, 2003). However, it is recognized that relative to direct precipitation reaching the land surface, stemflow is both slower and more gradual, yet because of direct connectivity with preferential flow paths along root systems and macropores, stemflow can also result in deeper infiltration and better drainage (Band et al., 2014; Ghestem et al., 2011). Thus while stemflow may account for some of the rainfall differences due to canopy interception, results from our hydrologic monitoring suggest that effective rainfall intensity and storm totals still increase with landslide disturbances, leading to enhanced hillslope wetness that also promotes instability.

Whereas these aforementioned impacts of vegetation removal on hillslope hydrology and stability are qualitatively well understood, here we examine their quantitative influence on hillslope hydrology in combination with other newly identified impacts of landslide disturbances on subsurface properties. Although dilation is commonly associated with landsliding, our observations and measurements suggest that compaction of the porous media structure was more prevalent at LS. Macropores and soil structures not only contribute to greater unsaturated storage and reduce effective water-retention properties (Mirus, 2015), but may also allow efficient lateral drainage (Sidle et al., 2001) and rapid transmission through the unsaturated zone to sometimes surprisingly great depths (Mirus & Nimmo, 2013; Nimmo, 2012). Similarly, matrix porosity and saturated hydraulic conductivity are fundamental properties that define storage and subsurface flow capacity. A reduction in soil structures and macropores, as well as matrix-saturated hydraulic conductivity and porosity at LS together contribute to the observed reduction in hillslope drainage and increased saturation levels over both event and seasonal time scales. Although enhanced hillslope drainage from higher hydraulic conductivity and macropores limits the landslide potential at VH, in some settings dead-end or blocked macropore networks can also promote locally elevated pore pressures that may actually trigger landsliding (Sharma, 2015). Therefore, further investigation of how macropores affect slope stability along the coastal bluffs of Puget Sound is needed.

Our field observations also suggest that the changes in topographic slope in upper portions of the LS in combination with the vegetation removal and altered hydraulic properties promote two types of episodic failures. First, the over-steepened scarps of glacial deposits at locations LS.1 and LS.2 are prone to topples and falls, which may be exacerbated by the removal of root structures (Figure 4 and supporting information Figures S4–S6) and therefore loss of any associated root strength and apparent cohesion. Second, the gently sloping portions of the slump blocks are prone to saturation (Figure 8) and shallow rapid earth flows (supporting information Figures S1, S7, and S8), likely due in large part to restricted drainage from the

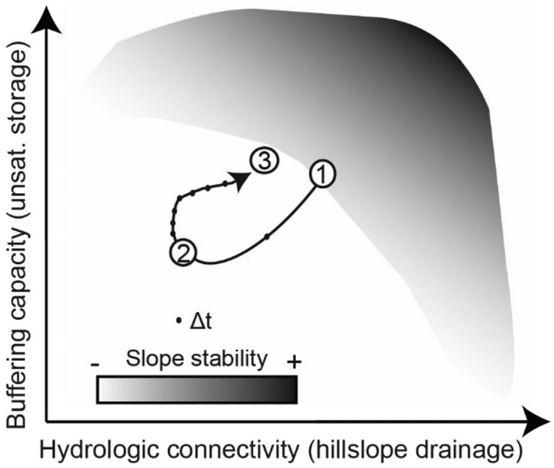

Figure 10. Hypothetical disturbance-recovery trajectory (see Ebel & Mirus, 2014) adapted for Puget Sound coastal bluffs to conceptually illustrate hydrologic connectivity, buffering capacity, and slope stability for a hillslope through time (Δt). First, a failure-prone vegetated hillslope (1) experiences an abrupt initial failure, which reduces hillslope drainage and unsaturated storage capacity, thereby promoting landslide reactivation (2). The failed hillslope (2) subsequently follows a slow recovery trajectory involving revegetation, bioturbation, and pedogenesis, which gradually increases hillslope drainage and unsaturated storage to promote slope stability (3).

decreased slope and altered hydraulic properties (Figures 2 and 5). Thus, our empirical evidence suggests that when combined with other disturbance-related changes to the porous media, gently sloping portions of the landslide deposit result in decreased subsurface flow-path connectivity and reduced drainage.

It is likely that the coastal bluffs along Puget Sound and other geologically similar settings undergo continuous cycles of landslide disturbance and recovery as illustrated conceptually in Figure 10. Despite the impacts of landslide disturbance documented in this study, the processes of revegetation, bioturbation, and pedogenesis collectively reduce the likelihood of landslide reactivation by increasing root reinforcement and enhancing hillslope drainage, thereby gradually stabilizing the steep coastal bluffs in the region. Continued monitoring of the recovery at LS as well as a broader investigation of other recovering landslides in the region could inform a quantitative framework for the conceptual model proposed in Figure 10. For example, Mirus and Loague (2013) used a figure qualitatively depicting the environmental controls on runoff generation mechanisms proposed by Dunne (1978) and data from four experimental catchments as a basis for developing quantitative nondimensional rate/storage ratios and generalized runoff generation thresholds. These same nondimensional ratios for hydraulic conductivity versus rainfall intensity (rate) and unsaturated storage versus rainfall depth (storage) could be applied to represent the x- and y-axes in Figure 10, respectively. The stability field could be quantified using a limit-equilibrium factor of safety approach (e.g., Lu & Godt, 2013). As with Mirus and Loague (2013), thresholds could be identified with numerical experiments rather than exhaustive fieldwork, but quantifying the time-step of recovery trajectories would be more complicated.

Although our empirical study examines only one landslide and one stable hillslope with similar characteristics, it is a useful point of comparison. Furthermore, it is unclear which of the individual factors associated with vegetation removal, disturbed subsurface properties, and topographic changes dominate, but prior research suggests that reduced root strength, evapotranspiration, and hydraulic conductivity are among the more likely contributors to enhanced landslide potential (Ebel et al., 2008; Roering et al., 2003; Schmidt et al., 2001; Ziemer, 1981). Regardless of which factor dominates slope stability, our results suggest their combined influence should be considered when assessing landslide hazards in settings where reactivation is possible. To evaluate the uniqueness of our hydrologic monitoring results from only one pair of hillslopes and quantify the importance of each factor on the stability of recent landslides in the Seattle area, we could use a virtual experiment approach with physically based numerical models (e.g., Ebel et al., 2008; McGuire et al., 2016; Mirus et al., 2007) informed by the measured hydraulic and geotechnical properties, as well as other observed differences between recent landslides and similar, but currently stable hillslopes. Preliminary efforts to apply a coupled hydromechanical model of variably saturated water flow and slope stability for these two hillslopes has highlighted some uncertainties in evapotranspiration, apparent root cohesion, and the deeper groundwater flow system (Mirus et al., 2017). In particular, additional fieldwork and/or sensitivity analyses are needed to establish whether our observed piezometer responses are due to an elevated regional groundwater table or the development of a near-surface zone of perched saturation.

We observed considerable heterogeneity within the glacial deposits exposed in headscarps at LS and also in an incised channel at VH (see also Mirus et al., 2016a), which could influence subsurface hydrologic response and may have played a role in the initial landslide at LS. However, these heterogeneities and the observed differences in colluvial deposits at VH compared to LS are at finer scale resolution than has been considered in previous landslide hazard assessments for the area (e.g., Baum et al., 2010). One critical implication of our findings relates to the potential need for improving hazard assessments to distinguish between the likelihood of a relatively new slope failure compared to the increased chances of reactivating recent landslide deposits. For example, the common practice of using soil texture to inform estimates of hydraulic properties for slope stability assessments with hydromechanical modeling (e.g., BeVille et al., 2010), may not be appropriate in situations where previous landslide deposits are concerned and could result in inaccurate estimates of porosity and hydraulic conductivity. Similarly, the use of universal rainfall thresholds to quantify landslide potential over large areas (Baum & Godt, 2010) might need to be adapted for environments where mobilized materials are not completely evacuated from the steep slopes to develop lower thresholds for landslide reactivation following widespread landsliding. In settings such as the coastal bluffs of Puget Sound, where disturbance impacts conspire to alter the hillslope hydrology, traditional methods for assessing landslide susceptibility may result in underestimates of the potential hazard for previously failed slopes. Given that only three occurrences of landslide reactivation at La Conchita, Oso, and West Salt Creek have collectively been responsible for at least 56 fatalities in the U.S. in just under a decade (i.e., 2005–2014), incorporating the impacts of landslide disturbance could improve the accuracy of hazard assessments and possibly reduce loss of life.

5. Conclusions

Topographic factors and vegetation removal are known to contribute to the recurrence of different types of landslides within a given area, but the role of landslide disturbance on hillslope hydrology and stability has remained largely unexplored. Here we propose an additional mechanism for landslide hazard persistence within this context of disturbance hydrology, and provide supporting evidence from a pair of hillslopes in Puget Sound, Washington, where steep coastal bluffs of unconsolidated glacial deposits are prone to shallow infiltration-induced landsliding. Overall, our site characterization and monitoring for a recent landslide and a control hillslope suggests that in such settings, where the failed material is not completely evacuated from the slope, the disturbances to vegetation and subsurface properties associated with landslides promote hydrologic conditions that contribute to extended and elevated landslide susceptibility. Although our largely empirical analysis does not quantify the relative importance of individual components of landslide disturbances on landslide hazard persistence, we do offer a compelling explanation for why landslides in certain settings recur where they have already happened in the past. Namely, landslide disturbances not only reduce apparent cohesion due to loss of root strength, but also reduce hillslope drainage and unsaturated storage capacity, both of which contribute to increased potential for landslide recurrence.

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